Predicting Chronic Kidney Disease from Clinical and Laboratory Measurements

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Introduction and hypotheses

Chronic kidney disease (CKD) is a long-term condition in which kidney function gradually declines. Because the kidneys filter waste products from the blood, regulate electrolytes, and maintain fluid balance, delayed identification of CKD can lead to serious health consequences. In clinical settings, physicians often evaluate CKD risk using routine measurements such as blood urea, serum creatinine, hemoglobin, blood pressure, albumin, diabetes status, and hypertension status. These measurements are relatively accessible compared with more invasive or expensive diagnostic procedures, so they are well suited for a data analysis project asking whether routine clinical variables can help distinguish CKD from non-CKD patients.

The real-world problem addressed in this project is binary medical classification: given a patient's clinical and laboratory measurements, can we predict whether the patient has CKD? This is important because a reliable screening model could help prioritize patients for further medical evaluation. Our goal is not to build a deployable diagnostic tool, but to use supervised classification methods from COGS 109 to study whether the available predictors contain enough information to separate CKD and non-CKD cases.

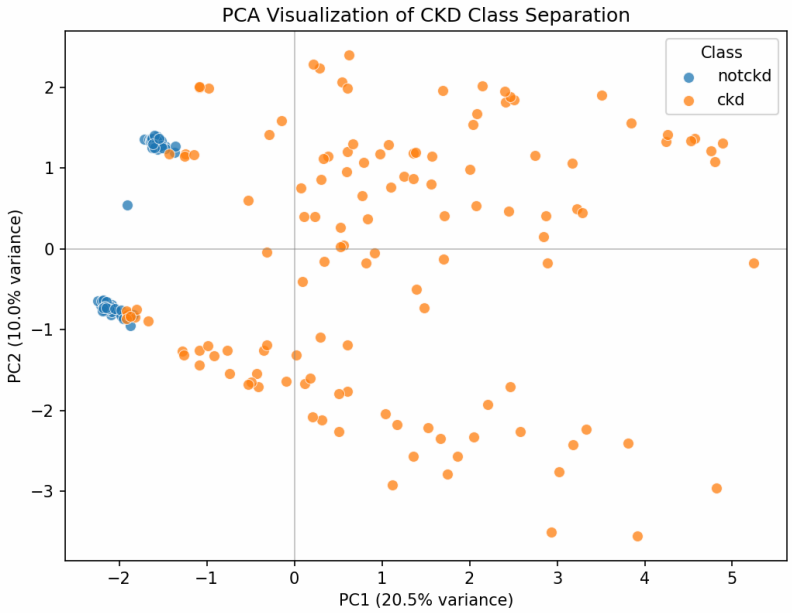
The dataset used in this project is a Kaggle mirror of the UCI Chronic Kidney Disease dataset. The original UCI dataset was donated in 2015 and is described as a classification dataset for predicting chronic kidney disease from clinical measurements collected from hospital data over approximately two months (Rubini, Soundarapandian, & Eswaran, 2015). The UCI repository describes the original dataset as having 400 instances and 24 features plus the class label. Our downloaded Kaggle version is a processed/binned version with 202 rows and 29 columns. After removing rows with invalid target labels and dropping variables that would leak the diagnosis (stage, affected, and grf), our final modeling dataset contains **n = 200 patients**, **p = 25 predictors**, and one binary target label. The target distribution is 128 CKD patients and 72 non-CKD patients.

The predictors include clinical and laboratory variables such as blood pressure, specific gravity, albumin, sugar, red blood cells, pus cells, blood glucose random, blood urea, sodium, serum creatinine, potassium, hemoglobin, packed cell volume, red blood cell count, white blood cell count, hypertension, diabetes mellitus, coronary artery disease, appetite, pedal edema, anemia, and age. Some variables are binary indicators, while others are represented as ordered or binned categories. Because the data are medical and partially discretized, possible noise sources include measurement variability, binning of continuous variables, missing or simplified clinical context, and the fact that the dataset is small.

Our main hypothesis is that kidney-related and general clinical predictors will be useful for CKD classification. Specifically, we expect variables related to kidney function and common CKD risk factors, including albumin, specific gravity, blood urea, serum creatinine,

hemoglobin/packed cell volume, hypertension, diabetes, pedal edema, and anemia, to help distinguish CKD from non-CKD patients. A secondary expectation is that more flexible models such as deep decision trees or very low-k K-nearest neighbors may achieve high training accuracy but show signs of overfitting when evaluated with cross-validation.

Initial exploratory analysis supported this hypothesis. The outcome classes were somewhat imbalanced: 64% of patients were labeled CKD and 36% were labeled non-CKD. Several clinically meaningful variables differed by class, including specific gravity, albumin, blood urea, serum creatinine, hemoglobin, hypertension, and diabetes.



PCA visualization of CKD class separation

PCA visualization of CKD and non-CKD patients. Each point represents one patient projected onto the first two principal components after preprocessing. The first two principal components explained 30.55% of the total variance, suggesting that the two classes have visible structure in the predictor space.

Methods - cross-validation

Before model selection, the dataset was split into an 80% training set and a 20% held-out test set. The split was stratified, meaning that the CKD and non-CKD class proportions were preserved in both sets. The training set contained 160 patients and the test set contained 40 patients. The training set had 102 CKD and 58 non-CKD observations; the test set had 26 CKD and 14 non-CKD observations.

Model selection was performed using **stratified 5-fold cross-validation** on the training set only. In 5-fold cross-validation, the training data are divided into five approximately equal-sized folds. For each fold, the model is trained on four folds and validated on the remaining fold. This process repeats five times so that every training observation is used once as validation data. The validation scores are then averaged to estimate how well the model generalizes to unseen data.

We used stratification so that each fold had approximately the same proportion of CKD and non-CKD cases as the full training set. We also shuffled the observations before creating folds and used a fixed random seed for reproducibility.

All preprocessing steps were placed inside the model pipeline during cross-validation. This means that imputation, standardization, and one-hot encoding were fit only on the training folds and then applied to the validation fold. This is important because preprocessing outside the cross-validation loop could leak information from validation observations into the training process.

The held-out test set was not used to choose the model family or hyperparameters. It was used only after model selection to estimate the final model's predictive performance. After selecting the model family and hyperparameter using training-set cross-validation, we evaluated the selected model once on the held-out test set. Finally, after model selection was complete, we refit the selected final model on the full dataset of 200 observations to obtain final parameter estimates, as required by the project rubric. The full-data refit was used for coefficient estimates, not for estimating predictive accuracy.

Methods - model approach

This project focuses on **prediction**, not causal inference. The target variable is whether a patient is classified as ckd or notckd, so the appropriate model class is supervised binary classification. The target was encoded as ckd = 1 and notckd = 0.

The modeling dataset excluded stage, affected, and grf because these variables are likely to encode information too directly related to the diagnosis or disease severity. Keeping them would risk data leakage, where the model appears accurate because it has access to information that would not be appropriate for real prediction. Numeric predictors were median-imputed and standardized. Categorical predictors were imputed using the most frequent category and then one-hot encoded, with the first category dropped to reduce redundancy.

We compared several classification methods covered in the course or closely related to course topics:

1. Logistic regression
2. K-nearest neighbors (KNN)
3. Decision tree classification
4. L1-regularized and L2-regularized logistic regression

Logistic regression is appropriate because it produces an interpretable linear decision boundary and coefficient estimates for each feature after preprocessing. KNN is useful as a nonparametric comparison model because it classifies a patient based on nearby patients in feature space. Decision trees provide a more flexible nonlinear model that can capture threshold-like decision rules. Comparing these models allows us to evaluate whether a simple regularized linear classifier is sufficient, or whether more flexible models improve predictive performance.

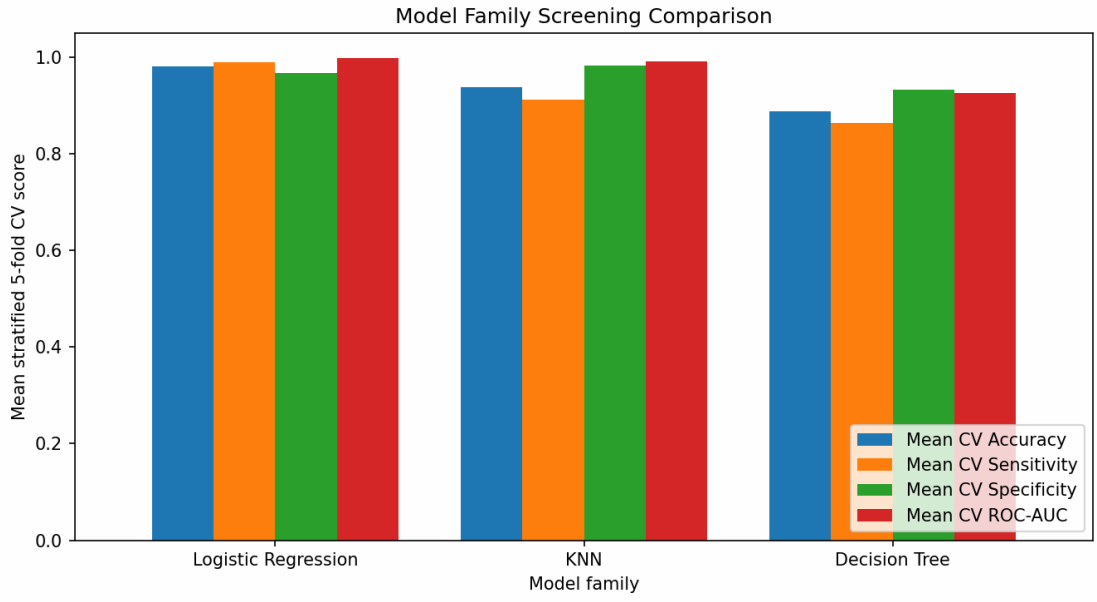
We also used principal component analysis (PCA) as an exploratory dimensionality-reduction method. PCA was not used as the final classifier. Instead, it was used to visualize whether CKD and non-CKD patients separate in a lower-dimensional representation of the predictor space.

This connects to the course topic of dimensionality reduction and helps interpret why the classification task may be relatively easy for the models.

We evaluated model performance using accuracy, sensitivity, specificity, ROC-AUC, and the confusion matrix. Accuracy measures the overall fraction of correct predictions. Sensitivity measures the fraction of CKD patients correctly identified as CKD, while specificity measures the fraction of non-CKD patients correctly identified as non-CKD. In this medical screening context, sensitivity is especially important because false negatives would mean missing patients who may have CKD. Specificity is also important because false positives could lead to unnecessary concern or follow-up testing.

Results - model selection

We first compared three model families using stratified 5-fold cross-validation on the training set: logistic regression, KNN, and decision tree classification. Logistic regression had the strongest overall cross-validation performance, with mean CV accuracy of 0.9812 and mean CV ROC-AUC of 0.9983. KNN also performed well but had lower mean CV accuracy of 0.9375. The decision tree model had the lowest screening performance, with mean CV accuracy of 0.8875 and mean CV ROC-AUC of 0.9263.



Model family comparison

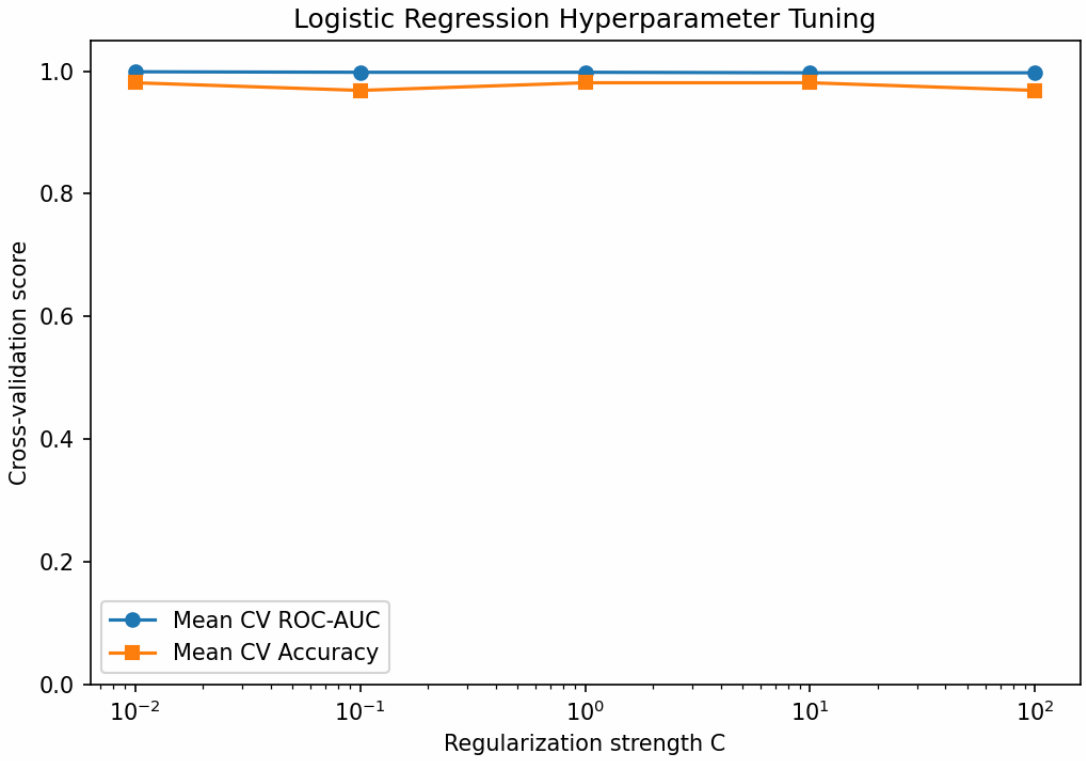
Model family screening comparison. Logistic regression, KNN, and decision tree classifiers were compared using mean stratified 5-fold cross-validation accuracy, sensitivity, specificity, and ROC-AUC.

Model	Mean CV Accuracy	Mean CV Sensitivity	Mean CV Specificity	Mean CV ROC-AUC
Logistic Regressio	0.9812	0.9905	0.9667	0.9983

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KNN	0.9375	0.9119	0.9833	0.9911
Decision Tree	0.8875	0.8643	0.9333	0.9263

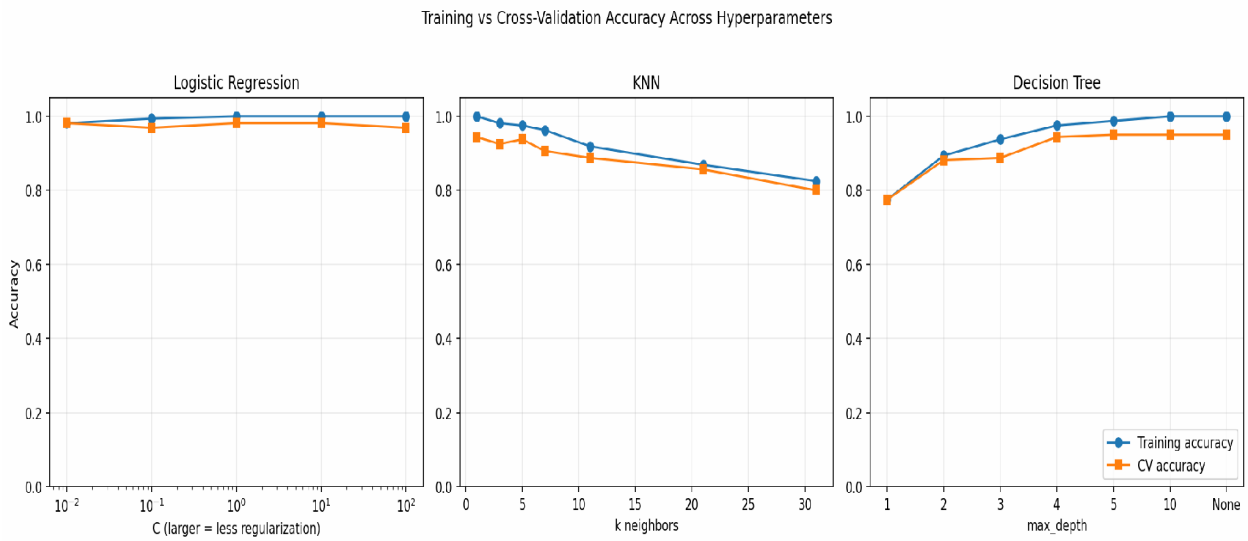
After logistic regression was selected as the best model family, we tuned its regularization parameter C. In logistic regression, smaller C corresponds to stronger regularization, while larger C corresponds to weaker regularization and greater model flexibility. The best value was C = 0.01, which had mean CV accuracy of 0.9812 and mean CV ROC-AUC of 0.9992.



Selected model tuning curve

Hyperparameter tuning for the selected logistic regression model. The tuning curve compares cross-validation performance across different values of C.

We also compared hyperparameter tuning curves across all three model families. This allowed us to examine model flexibility and overfitting. KNN with k = 1 achieved perfect training accuracy but lower CV accuracy of 0.9437, indicating higher variance. Decision trees with depth 10 or unlimited depth also achieved perfect training accuracy but only 0.9500 CV accuracy. In contrast, logistic regression with strong regularization (C = 0.01) had matching training and CV accuracy of 0.9812, suggesting better generalization and less overfitting.

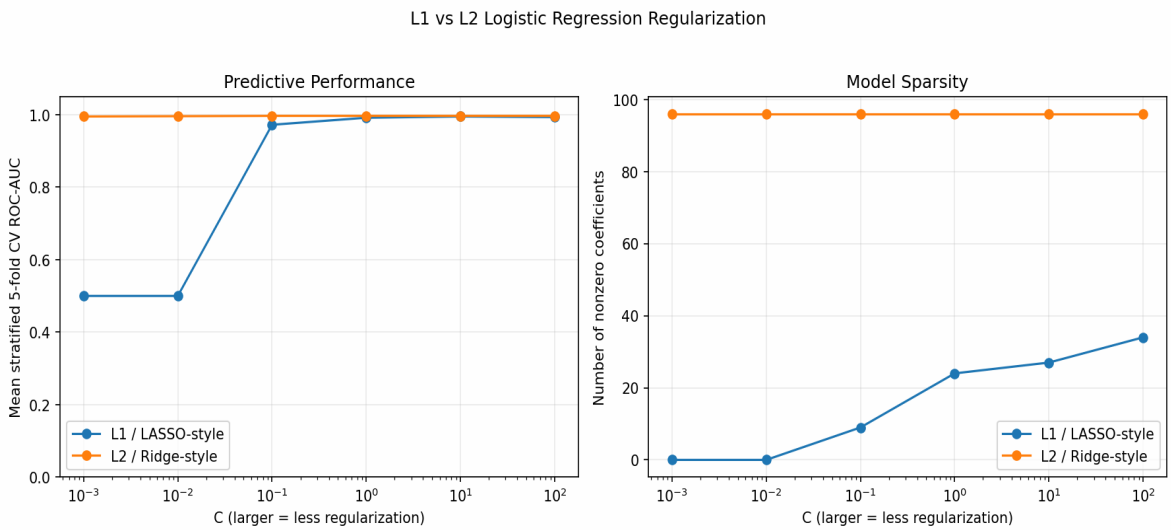


Training vs cross-validation performance across hyperparameters

Training vs cross-validation accuracy across hyperparameters. Larger gaps between training and CV accuracy indicate possible overfitting. KNN with small k and deeper decision trees show larger train-CV gaps than the selected regularized logistic regression model.

Model family	Hyperparameter	Value	Training accuracy	CV accuracy	Train - CV accuracy
Logistic Regression	C	0.01	0.9812	0.9812	0.0000
Logistic Regression	C	100	1.0000	0.9688	0.0312
KNN	k	1	1.0000	0.9437	0.0563
Decision Tree	max_depth	None	1.0000	0.9500	0.0500

To connect the project more directly to regularization, we compared L1-regularized logistic regression (LASSO-style) and L2-regularized logistic regression (ridge-style). L1 regularization can shrink some coefficients exactly to zero, producing a sparse model, while L2 regularization shrinks coefficients toward zero without usually removing them completely. The best regularized model by CV ROC-AUC was L2 logistic regression with C = 10.0, which had CV ROC-AUC of 0.9967 and 96 nonzero coefficients. L1 models showed the expected sparsity pattern: with very strong regularization (C = 0.001 or C = 0.01), all coefficients were zero and performance dropped to chance-level ROC-AUC of 0.5000; as C increased, more coefficients entered the model and performance improved.



L1 versus L2 regularization

L1 vs L2 logistic regression regularization. The left panel compares predictive performance across values of C; the right panel shows the number of nonzero coefficients. L1 regularization produces sparse models, while L2 regularization keeps all coefficients nonzero but shrinks them.

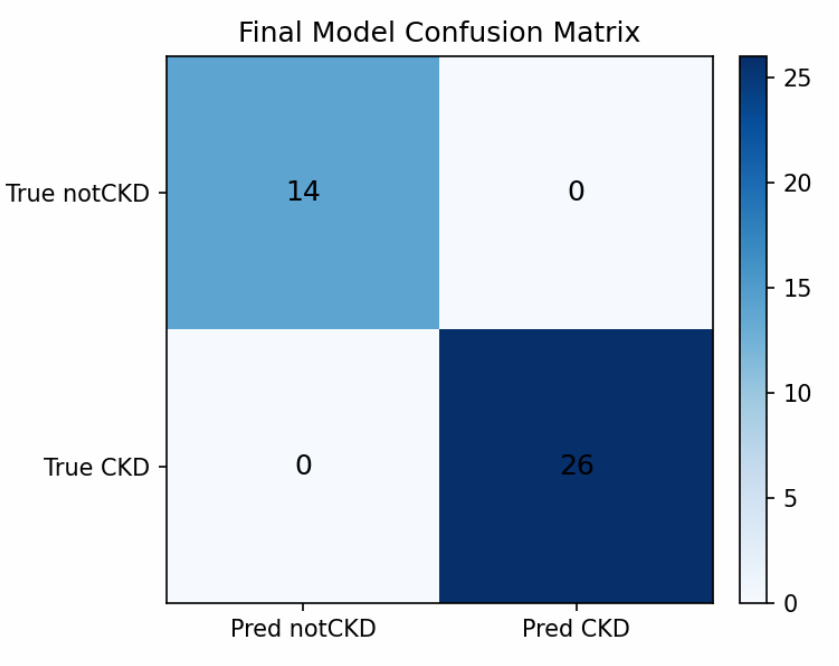
Overall, we chose logistic regression with C = 0.01 as the final model because it had the best CV ROC-AUC in the selected-family tuning step, high sensitivity and specificity, and little evidence of overfitting compared with more flexible KNN and decision tree settings.

Results - model estimation

The final selected model, logistic regression with C = 0.01, was trained on the 160-patient training set and evaluated on the 40-patient held-out test set. The model achieved perfect held-out test performance:

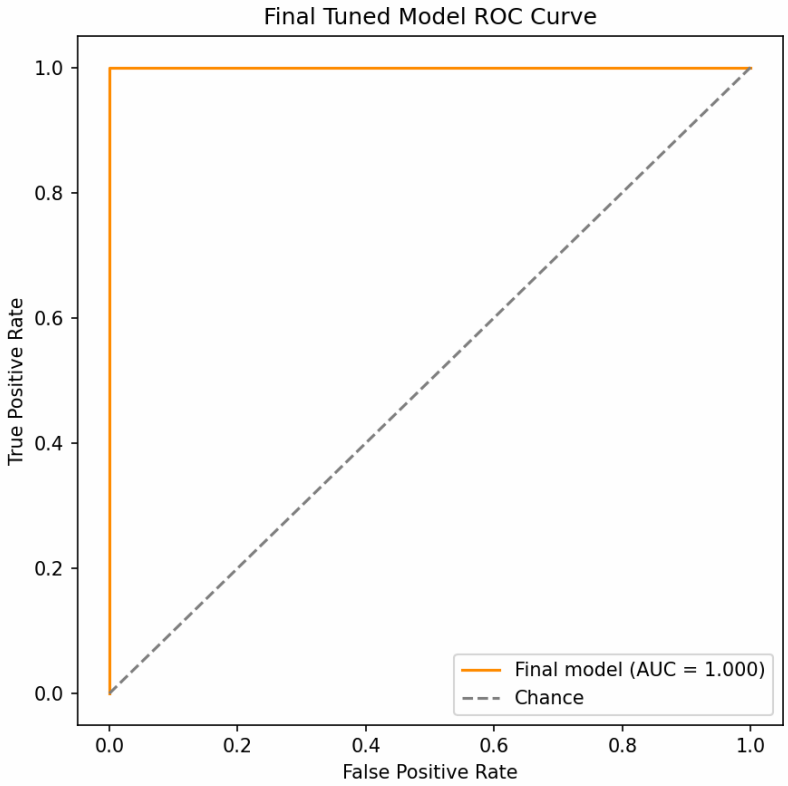
Metric	Held-out test value
Accuracy	1.0000
Sensitivity	1.0000
Specificity	1.0000
ROC-AUC	1.0000

The confusion matrix showed 14 true negatives, 26 true positives, 0 false positives, and 0 false negatives.



Final model confusion matrix

Final model confusion matrix on the held-out test set. The model correctly classified all 40 held-out test observations, including all 26 CKD patients and all 14 non-CKD patients.



Final tuned model ROC curve

ROC curve for the final tuned model on the held-out test set. The final logistic regression model achieved an ROC-AUC of 1.0000 on the held-out test set.

Because the rubric asks for final parameter estimates obtained using the full dataset, we refit the selected logistic regression model with C = 0.01 on all 200 observations after model selection was complete. The full-data logistic regression intercept was **0.9349**. The table below reports the largest full-data coefficient estimates by absolute value; the complete coefficient table is available in the modeling notebook.

Feature	Coefficient
Hypertension (htn)	0.2527
Diabetes mellitus (dm)	0.2350
Albumin < 0 (al_< 0)	-0.1486
Appetite (appet)	0.1403
Pus cell (pc)	0.1320
Pedal edema (pe)	0.1280
Specific gravity >= 1.023 (sg_>= 1.023)	-0.1214
Anemia (ane)	0.1165
Blood urea < 48.1 (bu_< 48.1)	-0.1136
Blood pressure limit (bp_limit)	0.1072

The signs of one-hot encoded categorical coefficients depend on the omitted reference category, so the coefficients should be interpreted as associations with the model's CKD log-odds relative to the reference levels after preprocessing, not as causal effects. Still, the largest coefficients are clinically plausible. Hypertension, diabetes, pedal edema, anemia, albumin, specific gravity, and blood urea are all related to CKD risk or kidney function. This supports our hypothesis that routine clinical measurements can distinguish CKD from non-CKD patients in this dataset.

One important limitation is that the held-out test accuracy was 100%. Although this is a strong result, it should be interpreted cautiously. Perfect performance can occur when a dataset is small, highly separable, discretized, or contains predictors that are very closely tied to the diagnostic label. It may also occur if there is remaining leakage that is not obvious. We attempted to reduce leakage by removing stage, affected, and grf, and by keeping preprocessing inside the cross-validation pipeline. However, because the test set had only 40 observations, a perfect score does not prove that the model would generalize perfectly to new hospitals, patient populations, or raw clinical records.

Conclusion and Discussion

The results support our main hypothesis. Clinical variables related to kidney function and common CKD risk factors were highly predictive of CKD status in this dataset. Logistic regression, KNN, and decision tree models all performed above chance, but logistic regression

had the strongest cross-validation performance and the smallest evidence of overfitting. The selected logistic regression model achieved mean CV accuracy of 0.9812 during model selection and 1.0000 accuracy, sensitivity, specificity, and ROC-AUC on the held-out test set.

The model comparison also supported our secondary expectation about model flexibility. Very flexible models, such as KNN with $k = 1$ and deep decision trees, achieved perfect training accuracy but lower cross-validation accuracy. This is a classic symptom of overfitting: the model fits the training observations extremely well, but its performance on validation data is lower. In contrast, the selected regularized logistic regression model had nearly identical training and cross-validation accuracy, suggesting a better bias-variance tradeoff for this dataset.

The regularization analysis further illustrated the relationship between model flexibility and performance. L1 logistic regression produced sparse models by setting some coefficients to zero, but extremely strong L1 regularization underfit the data and produced chance-level ROC-AUC. L2 logistic regression kept all coefficients but shrank them, producing consistently high cross-validation performance. This comparison shows why regularization is useful: it controls model flexibility and can improve generalization, especially when the number of encoded features becomes large relative to the number of observations.

The PCA visualization showed that CKD and non-CKD patients separate meaningfully in the first two principal components, even though those two components explained only 30.55% of the total variance. This suggests that the class structure is strong in the predictor space, which may help explain why several classifiers performed very well. However, PCA is unsupervised and does not use the class labels, so this visualization should be viewed as exploratory support rather than final evidence of predictive performance.

The main implication of this project is that routine clinical measurements can provide strong predictive information about CKD status in this dataset. From a research perspective, the most important next step would be external validation. In particular, the model should be tested on the original UCI CKD dataset or on an independent hospital dataset with the same variables. This would help determine whether the 100% held-out test accuracy reflects true generalizable structure or the small, processed nature of the current dataset. Future work could also compare performance before and after discretization, evaluate calibration of predicted probabilities, or test whether a smaller subset of clinically interpretable predictors can achieve similar performance.

Overall, this project demonstrates a complete supervised classification workflow: defining a binary prediction problem, cleaning and preprocessing clinical predictors, comparing multiple model classes with stratified cross-validation, examining overfitting through training-versus-CV curves, comparing L1 and L2 regularization, estimating final parameters using the full dataset, and interpreting the results with appropriate caution.

References

Rubini, L., Soundarapandian, P., & Eswaran, P. (2015). *Chronic Kidney Disease* [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5G020

UCI Machine Learning Repository. Chronic Kidney Disease dataset page. https://archive.ics.uci.edu/dataset/336/chronic+kidney+disease